

CorrBEV: Multi-View 3D Object Detection by Correlation Learning with Multi-modal Prototypes

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Abstract

Camera-only multi-view 3D object detection in autonomous driving has witnessed encouraging developments in recent years, largely attributed to the revolution of fundamental architectures in modeling bird's eye view (BEV). Despite the growing overall average performance, we contend that the exploration of more specific and challenging corner cases hasn't received adequate attention. In this work, we delve into a specific yet critical issue for safe autonomous driving: occlusion. To alleviate this challenge, we draw inspiration from the human amodal perception system, which is proven to have the capacity for mentally reconstructing the complete semantic concept of occluded objects with prior knowledge. More specifically, we introduce auxiliary visual and language prototypes, akin to human prior knowledge, to enhance the diminished object features caused by occlusion. Inspired by Siamese object tracking, we fuse the information from these prototypes with the baseline model through an efficient depth-wise correlation, thereby enhancing the quality of object-related features and guiding the learning of 3D object queries, especially for partially occluded ones. Furthermore, we propose the random pixel drop to mimic occlusion and the multi-modal contrastive loss to align visual features of different occlusion levels to a unified space during training. Our inspiration originates from addressing occlusion, however, we are surprised to find that the proposed framework also enhances robustness in various challenging scenarios that diminish object representation, such as inclement weather conditions. By applying our model to different baselines, i.e., BEVFormer and SparseBEV, we demonstrate consistent improvements.

1. Introduction

The camera-only solution currently stands out as one of the most captivating domains in both the industry and academia

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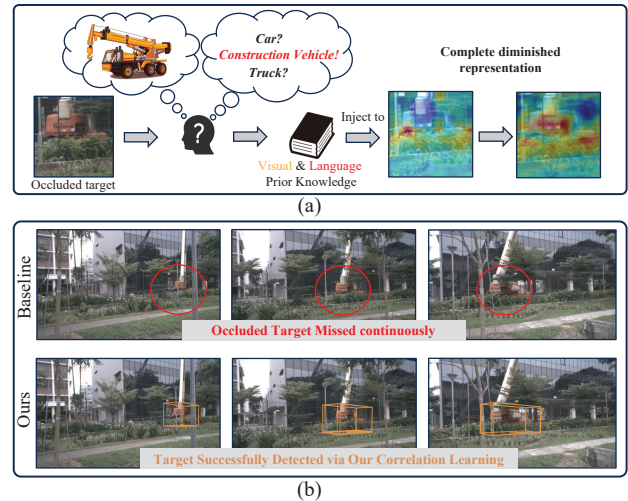


Figure 1. (a) The motivation of our proposed CorrBEV: injecting visual and language prior knowledge to complete dismissed representation in the real driving scene (taking the occluded target as an example); (b) The comparison of one baseline model (SparseBEV [28] here) and our method in detecting the occluded object.

of autonomous driving. 3D object detection from multi-view images, typically the initial phase of the solution, has witnessed significant progress in recent years. It is observed that most research efforts in 3D object detection are devoted to enhancing average performance. This inclination can be traced back to the original intention and design of evaluation benchmarks, in which metrics like NDS and mAP in nuScenes [2] are primarily proposed to average scores across all scenes and categories rather than emphasizing a special scenario. Consistent with this ethos, the foundational architectures of multi-view 3D object detection frameworks have undergone a remarkable evolution, transitioning from depth-estimated [14, 24] to depth-free [25, 30, 44], from single-frame [44] to multi-frame [13, 33], from dense-query [25] to sparse-query [26, 28].

However, the improvements in overall performance sometimes fall short of meeting the demands of developers, as critical agents often hold more significance for down-

stream planning systems but are easily overlooked by the average perception scores. For instance, the failure to detect a partially occluded pedestrian near the ego car has minimal impact on average performance but is crucial for ensuring safe autonomous driving. Recent works like RoboBEV [48] have reassessed the limitation of the current evaluation protocol and advocated for assessing methods under different scenarios of interest. Besides, methods designed for specified cases like long-range detection [15] are also introduced to address corner cases. While encouraged by the notable progress, we notice that the occlusion problem remains less explored. This motivates us to fill the gap in this work.

It’s easy to understand that there is no chance to localize the completely occluded target with solely single-frame knowledge. One effective approach that has been demonstrated to alleviate this challenge is the incorporation of more temporal information. Yet this is not the topic of our work. We primarily tend to enhance the detection quality of partially occluded objects with only single-frame clues. By doing so, our proposed method can seamlessly integrate with almost all existing methods regardless of their baseline architecture design. The resilience of our method will be proved in the later experiments. Then, *what is the inherent difficulty in detecting partially occluded objects?* Generally, occlusion diminishes the quality of learned visual representation as fewer discernible features are available compared with non-occluded objects. It’s akin to a chicken-and-egg scenario because detecting occluded objects necessitates sufficient contextual cues, yet such information is inherently limited due to occlusion. So, *could we introduce additional clues for occluded objects?*

We try to find the answer from the human vision and recognition systems, which are proven to have the capacity for mentally reconstructing the complete semantic concept of occluded objects with prior knowledge [17, 19]. This ability, known as amodal perception, enables us to assimilate integral information from both visible cues and imperceptible signals [36]. Inspired by this mechanism, we propose a novel plug-and-play architecture termed CorrBEV, which incorporates auxiliary semantic prototypes for occlusion reasoning. As an old saying goes, “A picture paints a thousand words”. One intuitive way is to supply richer visual priors from image templates that correspond with different human-defined object categories. Besides, human language, specifically the category name itself, encapsulates compact knowledge about the described object, which has already demonstrated effectiveness in improving vision tasks [9, 10, 29, 37, 52]. Therefore, it sounds reasonable to introduce language clues alongside the visual priors to enhance representation learning of partially occluded objects. We provide an example to illustrate this motivation in Fig. 1.

More specifically, our proposed framework consists of three core components. Firstly, the **Multi-modal Proto-**

type Generator provides prior knowledge for improving the representation quality of targets. We crop images with 2D ground-truth boxes in the training dataset and use a publicly available model to extract feature embeddings. The feature embeddings are then clustered based on visibility level¹ and category to form occlusion-aware visual prototypes. The language prototypes are obtained by transforming category names (e.g. pedestrian) to embeddings with pre-trained BERT [8]. Secondly, the **Correlation-guided Query Learner** injects the knowledge of multi-modal prototypes to the model, which is inspired by Siamese Object Tracking [1, 20, 55]. The depth-wise correlation operator transforms prototypes and backbone features into salient maps, which we find partially occluded objects can be better localized than that in 3D space. The decoder then generates 3D object queries with the salient maps and samples 2D features from both the backbone and the correlation feature in a dual way. Last but not least, during training, the pseudo-occlusion processor in our proposed **Occlusion-aware Trainer** randomly drops pixels in less-occluded objects, which can balance the distribution of objects with different occlusion levels. Besides, we also propose to align vision and language representation with contrastive loss [32]. Considering one language embedding as a cluster center, the alignment training forces the model to predict closer representation in the latent space for occluded and non-occluded objects of the same category.

To demonstrate the effectiveness and generality of the proposed method, we apply our model to both BEVFormer [25] (dense-query) and SparseBEV [28] (sparse-query). Without bells and whistles, our model improves the aforementioned two baselines for 2.6%/2.7% mAP and 2.6%/1.6% NDS on nuScenes [2] validation split, respectively. The advances are maintained in our introduced occlusion-aware evaluation. More surprisingly, while our approach mainly aims at mitigating the negative effects of occlusion, we observed that our method also enhances the robustness of baseline techniques in other challenging scenarios from RoboBEV [48]. The results highlight the resilience of our proposal, which can effectively address the diminished representation caused by various conditions encountered in the real world.

In summary, our contributions are four-fold: (i) We propose a plug-and-play model that incorporates auxiliary visual and language clues to improve the robustness of detection methods, especially on partially occluded objects. (ii) We innovatively introduce correlation learning to elegantly inject the knowledge of multi-modal prototypes into multi-view 3D detection models. (iii) We optimize the training pipeline of multi-view detection methods by proposing

¹In nuScenes [2], “visibility” is defined as the fraction of pixels of a particular annotation that are visible over the 6 camera feeds, grouped into 4 bins. Please refer to nuScenes [2] and Sec. 4 for more details.

random masking and multi-modal alignment to improve the capability of detecting occluded objects. (iv) We prove the effectiveness of our model on different baseline methods.

2. Related Work

Multi-view Camera-based 3D Object Detection. Benefiting from the intrinsic complementarity in different observation angles, recent year’s attention in camera-based 3D object detection is turning from a single-view to its multi-view counterparts. The related methods can be roughly categorized into two branches. The first one explicitly constructs dense BEV representations by lifting the 2D image features to 3D space. Specifically, the pioneering work LSS [35] performs view transformation with the predicted categorical depth distribution of each pixel. The follow-up method BEVDet [14] designs rich image-level and BEV-level augmentations in training. BEVDepth [24] and BEVStereo [23] further improve the quality of BEV features with explicit depth supervision. Another branch densely or sparsely projects 3D queries to 2D images for sampling and aggregating visual information. The pioneering work DETR3D [44] is inspired by its 2D parent DETR [4], which builds relations between sparse 3D object queries with 2D points in the images. BEVFormer [25] chooses to explicitly learn BEV representation. PETR [30] employs 3D position coding to generate 3D position-aware features. Sparse4D [26] and SparseBEV [28] can be regarded as the temporal version of DETR3D [44], which explores a fully sparse manner to aggregate multi-frame features.

As the recent consensus in practical application suggests, handling corner cases may be the last mile of perception systems. The research community also proposes different works to catch up with this trend. For instance, RoboBEV[48] tags the nuScenes with different challenging attributes for more sufficiently evaluating models. Far3D [15] and LR3D [50] pay attention to long-distance objects, which tends to expand the spatial range of BEV detection methods. OpenSight [53] proposes to leverage multi-modal LiDAR and camera data for open-vocabulary 3D object detection. In this work, we focus on handling occlusion cases which are essential for safe autonomous driving.

Handling Occlusion in 3D Object Detection. Occlusion presents an inevitable and critical challenge in autonomous driving, particularly in the realm of 3D object detection. Significant efforts have been made to address this issue through LiDAR-based [12, 40, 49] and camera-based [5, 18, 39, 51] solutions. In LiDAR-based approaches, a common strategy involves recovering hidden 3D information. For example, WYSIWYG [12] minimizes shape distortion due to occlusion by utilizing visibility information via a raycasting algorithm. BtcDet [49] incorporates a sub-network designed to estimate the complete occupancy of objects in occluded regions. Camera-

based methods tackle occlusion through modeling 2D occluded keypoints [39], reconstructing boxes using shape priors [18], predicting 2D depth maps [41], and estimating uncertainty [5]. Additionally, a widely recognized strategy for mitigating the adverse effects of occlusion involves the fusion of temporal information [26, 31, 33, 42]. However, temporal models may lose efficacy for moving objects due to the challenges of accurate motion compensation. Moreover, the ambiguous context of occluded targets and their foreground objects can diminish the quality of temporal learning. In camera-based multi-view 3D object detection, we observed that the occlusion challenge has received limited attention. Previous work OccluBEV [46] predicts 3D voxel segmentation masks based on labels derived from LiDAR points. Differently, we propose a general framework that leverages vision-language multi-modal information to enhance diminished object representations caused by occlusion and other challenging driving environments, without relying on additional LiDAR data.

3. Method

This section details our proposed framework CorrBEV, as shown in Fig. 2. We first delineate the process of generating visual and language prototypes in Sec. 3.1. Then, in Sec. 3.2, we propose correlation learning to use the prior knowledge of multi-modal prototypes. Finally, Sec. 3.3 introduces the designed occlusion-aware trainer.

3.1. Multi-modal Prototype Generator

As aforementioned, insufficient contextual cues introduce an inherent challenge in detecting occluded objects. In this work, we propose the Multi-modal Prototype Generator to mitigate this problem by imitating human imagination ability with auxiliary prior knowledge beyond the observations (images). Specifically, for learning visual prototypes, we offline crop 2D object templates in the bounding boxes and use DeViT [54] to transform these templates to feature embeddings $\mathbf{P}_v \in \mathbb{R}^{K \times N \times D}$, where K and N denote the number of samples and categories, respectively. D is the number of channels. As we primarily want to alleviate the occlusion problem, we split the prototypes based on the visibility label of each ground-truth box and then average the K samples, obtaining $\mathbf{P}_{vo} \in \mathbb{R}^{(N \times M) \times D}$. M is the number of visibility levels ($N = 10$, $M = 4$ in nuScenes [2]). For the language prototypes, we use the pre-trained BERT [8] to extract high-level semantic information in object category names, obtaining $\mathbf{P}_l \in \mathbb{R}^{N \times D}$. To adapt the prototypes to the detection task, we fine-tune the text encoder during training. Since we use the relatively light BERT encoder, rare training costs would be introduced (see experiments). The visual prototypes are offline generated, therefore the image encoder is frozen in training. Besides, training the image encoder is not a wise decision considering its cost.

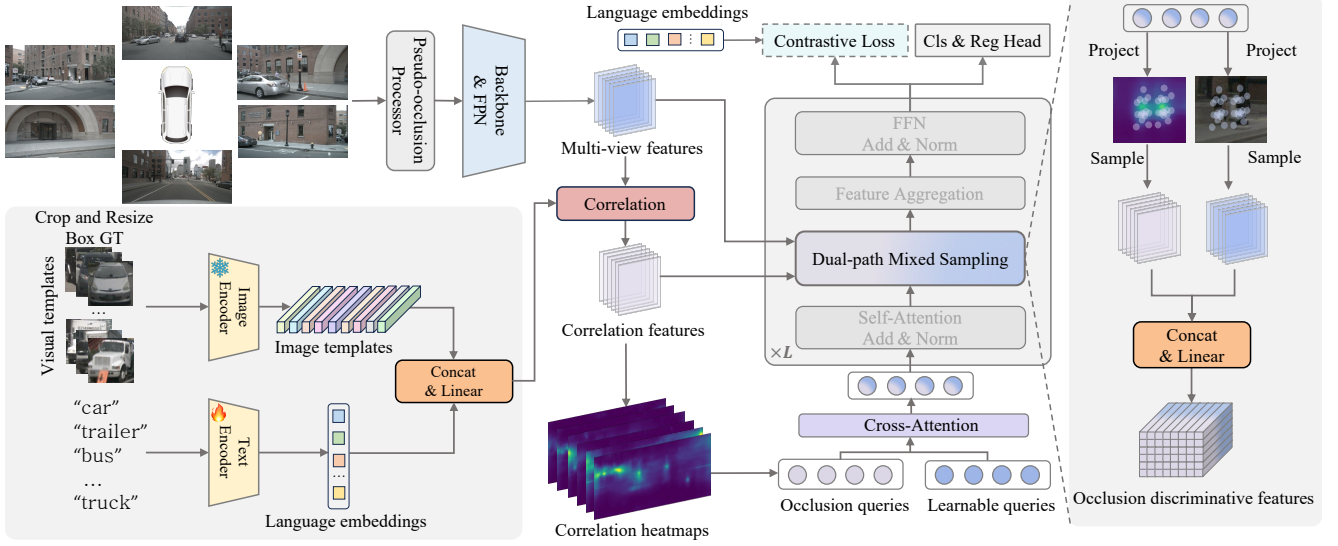


Figure 2. Overall architecture of the proposed CorrBEV for multi-view 3D object detection. The vision-language prototypes provide prior knowledge to improve the capability of detecting occluded objects.

In our work, we innovatively propose a simple way to fine-tune the visual prototypes. In specific, we initialize a tensor of trainable parameters (nn.Embedding in PyTorch [34]) with the raw visual embedding. Finally, we concatenate and squeeze the visual and language prototypes along the channel dimension by broadcasting, obtaining $\mathbf{P} \in \mathbb{R}^{(N \times M) \times D}$.

3.2. Correlation-guided Query Learner

The next step is to fuse the multi-modal prototypes with baseline models. There are three predefined goals: 1) guarantee the efficiency; 2) enhance the feature quality, especially the partially occluded objects; 3) guide the detection of partially occluded objects. In this work, we introduce the correlation learner to satisfy all the requirements.

For efficiency, we introduce the depth-wise correlation from Siamese Object Tracking [1, 56] to fuse multi-modal prototypes with the backbone features $\mathbf{F}_b \in \mathbb{R}^{H \times W \times D}$. H and W represent the height and width of the feature. The implementation of this operator equals to depth-wise convolution in modern deep-learning frameworks². Hence, the efficiency (our first goal) is guaranteed, while achieving the correlation feature $\mathbf{F}_{corr} \in \mathbb{R}^{H \times W \times (N \times M \times D)}$,

$$\mathbf{F}_{corr} = \text{Conv}_{1 \times 1}(\mathbf{P}, \mathbf{F}_b), \quad (1)$$

where $\text{Conv}_{1 \times 1}$ is the 1×1 convolution operator which takes the fused multi-modal prototypes $\mathbf{P}$ as the convolution kernel. We reduce the feature dimension of $\mathbf{F}_{corr}$ to $\frac{D}{4}$ for efficiency. Correlation is an operator that was initially designed to find the most similar objects with exemplars [1] in an image. The depth-wise version extends the original

²Modern deep-learning frameworks implement “convolution” as “correlation” because of its efficiency and similar functionality [34].

correlation operator to multi-dimension [20], aiming at enhancing target-related features meanwhile decreasing the response of unrelated backgrounds (our second goal).

Considering the discriminative semantic properties within the correlation feature, we try to take its advantage for occlusion reasoning from two aspects. Firstly, we propose Object Query Initializer to guide the generation of object candidates, *i.e.*, 3D object queries in DETR-like decoder [44], with the correlation feature. The motivation is inspired by our observation that it’s easier to localize partially occluded objects in 2D images than in 3D BEV space, since the feature sampling of 3D object queries may unexpectedly mix an occluded target with its foreground object. Specifically, we predict confidence maps $\mathbf{H}_{vis} \in \mathbb{R}^{H \times W \times N \times M}$ for each visibility and category with correlation feature $\mathbf{F}_{corr}$. We extract the embeddings from the correlation feature at the top k positions of the confidence maps (k is the number of 3D object queries). The embeddings are regarded as target-aware object queries $\mathbf{Q}_t \in \mathbb{R}^{k \times D}$,

$$\mathbf{Q}_t = \{\mathbf{F}_{corr}^i \mid i \in \text{argmax}_k \mathbf{H}_{vis}\}, \quad (2)$$

where i is the pixel index. The target-aware queries $\mathbf{Q}_t$ are fused with learnable queries $\mathbf{Q}_{learn}$ in the baseline models by cross-attention,

$$\mathbf{Q} = \text{CrossAttn}(q = \mathbf{Q}_{learn}, k = \mathbf{Q}_t, v = \mathbf{Q}_t). \quad (3)$$

The number of object queries $\mathbf{Q}$ is the same as that in baseline models, guaranteeing fair comparisons.

Secondly, we try to construct object-aware 3D object representations with the correlation feature. As aforementioned, the diminished 2D features of partially occluded objects would degrade the learning effectiveness of 3D object

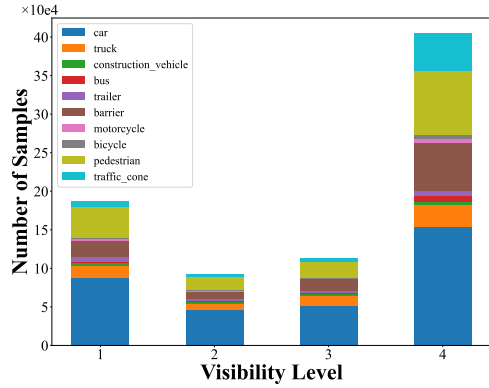


Figure 3. The distribution of different visibility levels in nuScenes.

queries. To alleviate the occlusion ambiguity, we propose Dual-path Mixed Sampling to guide the learning of 3D object queries. In specific, the 3D object queries sample features by projecting the predefined 3D sampling points p to the 2D feature planes (*i.e.*, the backbone feature $\mathbf{F}_b$ and the correlation feature $\mathbf{F}_{corr}$), which is guided by predicting sampling offsets Δp . Then, the 3D object queries $\mathbf{Q}$ are updated by aggregating the sampled features,

$$\mathbf{Q}_{up} = \text{Linear} \left[\frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} \sum_{j=1}^{N_{ref}} \mathbf{F}_b^{i,j}(p_{ij} + \Delta p_{ij}); \right. \\ \left. \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} \sum_{j=1}^{N_{ref}} \mathbf{F}_{corr}^{i,j}(p_{ij} + \Delta p_{ij}) \right] + \mathbf{Q}, \quad (4)$$

where Linear represents the linear layer, $[\cdot; \cdot]$ indicates the concatenation along channel dimension. N_{ref} is the total reference points for each 3D object query. $|\mathcal{V}|$ is the number of surrounding cameras. With the proposed method, our correlation-based query learner can effectively guide the detection of partially occluded objects (our third goal).

3.3. Occlusion-aware Trainer

Pseudo-occlusion Processor. As shown in Fig. 3, the number of objects with different visibility levels is obviously unequal, which poses a challenge to optimize targets in minorities. Motivated by data augmentation in 2D tasks [57], we introduce a pseudo-occlusion processor (P2) to balance the distribution of objects with different visibility levels. Specifically, the proposed P2 randomly drops pixels in the 2D bounding boxes of non-occluded objects (*i.e.*, with visibility value of 3 or 4 in nuScenes [2]) and replaces the dropped pixels with mean values of the region.

Contrastive Semantic Alignment. Contrastive learning was generally used in self-supervised learning [6] and multi-modal alignment [38]. We introduce contrastive learning in this work to: 1) intra-class: align the representa-

tion of objects with different occlusion levels to one unified space (*i.e.*, language), therefore enhancing the classification accuracy of partially occluded objects, 2) inter-class: enlarge the representation distinguishability of different categories. In particular, we follow previous works [21, 29] applying dot product between each query with language prototypes to predict logits and compute focal loss for each query.

4. Experiments

4.1. Datasets and Metrics

Following our baselines [25, 28], we conduct experiments on the nuScenes [2] dataset. The benchmark comprises 1000 driving scenes and is split into 700/150/150 for training/validation/testing. For 3D object detection, there are about 1.4M annotated 3D bounding boxes from 10 classes. Alongside coordinates and scale labels of the 3D bounding box, the dataset also provides object visibility levels. The visibility levels are officially divided into four ranges, *i.e.*, 0-40%, 40-60%, 60-80%, and 80-100%. Since occlusion is the key factor for low-visibility samples, we regard the performances on four visibility levels as the indicator of detection robustness under different occlusion conditions.

For evaluating and comparing the detection performances of different methods, we adopt the mean average precision (mAP) and nuScenes detection score (NDS) as primary metrics, as well as five True Positive (TP) metrics, including mean Average Translation Error (mATE), mean Average Scale Error (mASE), mean Average Orientation Error (mAOE), mean Average Velocity Error (mAVE), and mean Average Attribute Error (mAAE). Additionally, we report the recall of four visibility levels to evaluate the detection capability on partially occluded objects.

4.2. Implementation Details

As mentioned before, the proposed method can be easily plugged into recent multi-view 3D object detection methods. To prove the generality of our method, we apply it to both sparse-query based SparseBEV [28] and dense-query based BEVFormer [25]. For fair comparisons, we adopt the same training settings with the baselines. All experiments are performed with 8 NVIDIA GeForce RTX 3090 GPUs.

4.3. State-of-the-art Comparison

Evaluation of Overall Performance. To evaluate the proposed CorrBEV, we apply our method to BEVFormer_{small} [25] and SparseBEV [28], dubbed CorrBEV_{fm} and CorrBEV_{sp}, respectively. The results are presented in Tab. 1 and Tab. 2. Compared with the baseline BEVFormer_{small} [25], our CorrBEV_{fm} obtains considerable gains of 2.6%/3.6% mAP and 2.6%/2.0% NDS on the validation and test splits of nuScenes, respectively. The advancement of our method is also preserved for sparse-query

Table 1. Performance comparison on validation split of nuScenes [2]. “Res50” and “Res101” denote ResNet50 [11] and ResNet101 [11], respectively. [†] denotes without using historical frames. “-” indicates the checkpoint is not provided by the corresponding method.

Method	Backbone	mAP [†]	NDS [†]	mATE [↓]	mASE [↓]	mAOE [↓]	mAVE [↓]	mAAE [↓]	Vis ₁ Recall [†]	Vis ₂ Recall [†]	Vis ₃ Recall [†]	Vis ₄ Recall [†]
BEVDet [14]	Res50	29.8	37.9	0.725	0.279	0.589	0.860	0.245	34.7	46.0	56.0	64.8
BEVDet4D [13]	Res50	32.2	45.7	0.703	0.278	0.495	0.354	0.206	36.2	47.8	55.5	64.6
BEVDepth [24]	Res50	35.1	47.5	0.639	0.267	0.479	0.428	0.198	34.8	46.9	56.2	66.8
BEVStereo [23]	Res50	37.2	50.0	0.598	0.270	0.438	0.367	0.190	35.1	46.4	57.6	66.6
PETR [30]	Res50	31.7	36.7	0.839	0.280	0.615	0.953	0.232	45.6	58.6	62.7	69.8
PETRv2 [31]	Res50	34.9	45.6	0.700	0.275	0.580	0.437	0.187	-	-	-	-
HoP [58]	Res50	40.0	51.0	0.607	0.275	0.515	0.286	0.216	49.8	59.3	67.0	74.6
SOLOFusion [33]	Res50	43.7	53.8	0.567	0.274	0.511	0.252	0.181	48.8	59.1	66.4	71.8
StreamPETR [42]	Res50	45.0	55.0	0.569	0.267	0.413	0.265	0.196	59.9	69.3	73.0	77.9
RayDN [27]	Res50	46.9	56.3	0.579	0.264	0.433	0.256	0.187	60.5	70.4	75.0	78.3
SparseBEV [28]	Res50	44.8	55.8	0.581	0.271	0.373	0.247	0.190	60.4	70.5	74.9	80.0
CorrBEV _{sp} (ours)	Res50	47.5	57.4	0.598	0.253	0.335	0.246	0.200	69.1	77.6	83.0	87.4
SparseBEV [28]	Res101	50.1	59.2	0.562	0.265	0.321	0.243	0.195	69.1	77.7	80.3	82.9
CorrBEV _{sp} (ours)	Res101	51.7	60.6	0.554	0.247	0.298	0.241	0.189	72.7	82.3	84.6	88.4
BEVFormer [†] _{small} [25]	Res101	33.2	40.5	0.786	0.282	0.438	0.862	0.221	46.4	58.4	64.1	69.9
CorrBEV [†] _{fm} (ours)	Res101	35.2	42.3	0.751	0.275	0.416	0.853	0.211	47.9	59.9	64.2	70.8
BEVFormer _{small} [25]	Res101	37.0	47.9	0.721	0.280	0.407	0.436	0.220	50.7	60.6	65.2	72.4
CorrBEV _{fm} (ours)	Res101	39.6	50.5	0.689	0.279	0.368	0.407	0.191	51.3	62.8	67.0	72.3

Table 2. Performance comparison on the nuScenes test split.

Method	Backbone	mAP [†]	NDS [†]	mATE [↓]	mASE [↓]	mAOE [↓]	mAVE [↓]	mAAE [↓]
FCOS3D [43]	Res101	35.8	42.8	0.690	0.249	0.452	1.434	0.124
PGD [7]	Res101	38.6	44.8	0.626	0.245	0.451	1.509	0.127
PETR [30]	Res101	37.0	45.5	0.647	0.251	0.433	0.933	0.143
UVTR [22]	Res101	37.9	48.3	0.731	0.267	0.350	0.510	0.200
PolarFormer [16]	Res101	41.5	47.0	0.657	0.263	0.405	0.911	0.139
M ² BEV [47]	Res101	42.9	47.4	0.583	0.374	0.254	0.376	1.053
BEVFormer _{base} [25]	Res101	44.5	53.5	0.631	0.257	0.405	0.435	0.143
FrustumFormer [45]	Res101	47.8	56.1	0.575	0.257	0.402	0.411	0.132
BEVFormer _{small} [25]	Res101	37.7	48.7	0.675	0.268	0.454	0.479	0.138
CorrBEV _{fm} (ours)	Res101	41.3	50.7	0.646	0.265	0.482	0.473	0.134
SparseBEV [28]	Res101	54.3	62.7	0.502	0.244	0.324	0.251	0.126
CorrBEV _{sp} (ours)	Res101	56.2	64.0	0.488	0.241	0.303	0.243	0.123

based method. It shows that with SparseBEV [28], our CorrBEV_{sp} gains improvements of 2.7% mAP and 1.6% NDS on the validation split. Notably, our module only introduces 7.48% additional training costs to the baseline SparseBEV (21.41 v.s. 19.92 GPU days on a single RTX 3090). And the impact on inference speed is small, with 18.4 FPS compared to the baseline’s 21.7 FPS. As shown in the table, our model can also be easily integrated into the single-frame based detector, the static BEVFormer[†]_{small} with the proposed model improves the baseline for a considerable 2.0% mAP. The results demonstrate the effectiveness and generalization of the proposed framework.

Evaluation of Occlusion Performance. Besides the enhancement of overall performance, we also show improve-

ments in detecting occluded objects with our method. Before detailing the results, we first describe an interesting observation. As shown in Tab. 1, the overall performance of SOLOFusion [33] clearly outperforms the baseline BEVFormer_{small} for 6.7% mAP. Yet surprisingly, the recalls on severely occluded conditions (Vis₁ and Vis₂) are pretty lower than BEVFormer_{small}. This proves our above claim that the improvements in overall performance and occlusion robustness may not be consistent. By applying our model to the two baseline frameworks, performances across all occlusion levels are consistently enhanced, which demonstrates the effectiveness of our design by introducing multi-modal prototypes to provide prior knowledge for occlusion reasoning. We surprisingly find that the severely

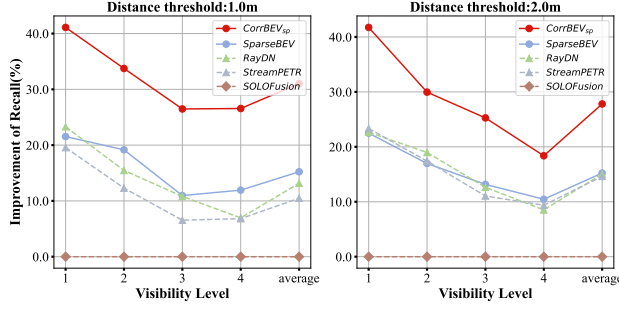


Figure 4. Relative improvement comparison of recall (SOLOFusion [33] serves as the reference method).

Table 3. Component-wise ablation on nuScenes validation split. CL denotes contrastive loss and P2 is Pseudo-occlusion Processor.

#	Prototype Type	CL	P2	mAP↑	NDS↑	Vis ₁ Recall↑	Vis ₂ Recall↑	Vis ₃ Recall↑	Vis ₄ Recall↑
①	-	-	-	44.8	55.8	60.4	70.5	74.9	80.0
②	✓	-	-	45.9	56.3	64.8	72.6	77.5	83.4
③	✓	✓	-	46.5	56.6	68.0	74.5	80.5	85.2
④	✓	✓	✓	46.9	57.1	68.3	77.1	82.6	86.8
⑤	✓	✓	✓	47.5	57.4	69.1	77.6	83.0	87.4

Table 4. Prototype fine-tuning ablation on nuScenes val split.

#	Prototype Training	mAP↑	NDS↑	Vis ₁ Recall↑	Vis ₂ Recall↑	Vis ₃ Recall↑	Vis ₄ Recall↑
①	-	45.9	54.8	62.2	71.1	76.5	82.4
②	✓	46.3	56.0	65.9	74.0	80.2	84.1
③	-	46.8	56.8	67.0	75.2	81.4	84.8
④	✓	47.5	57.4	69.1	77.6	83.0	87.4

occluded objects have the largest improvements compared with baseline SparseBEV (60.4% v.s. 69.1% recall in Vis₁). We also observe that for dense-query based methods like BEVFormer, the performance in high visibility (less occlusion) is close (72.4% v.s. 72.3% recall in Vis₄), while the conclusion is contrary for sparse-query based SparseBEV (80.0% v.s. 87.4% recall in Vis₄). This observation matches our intuition that dense visual feature sampling for non-occluded objects is sufficient enough for learning 3D representation, while additional prior knowledge can complete the 3D representation generated by sparse sampling. Moreover, we present the relative improvement of recall under different occlusion conditions in Fig. 4 for an intuitive comparison, where our model shows consistent improvements. Since the recall of partially occluded objects is crucial for safe autonomous driving (see Fig. 6), we believe our work could further benefit the downstream prediction and planning systems.

Evaluation of Performance in other Corner Cases. As stated above, our inspiration stems from the challenge of addressing occlusion. However, this is not the only goal. We also evaluate performance across various scenarios presented in RoboBEV [48], which similarly diminishes object representations. As illustrated in Fig. 5, applying our model to the baseline (BEVFormer_{small}) yields consistent improvements across all scenarios, particularly under *Snow* and *ColorQuant*. These encouraging results bolster

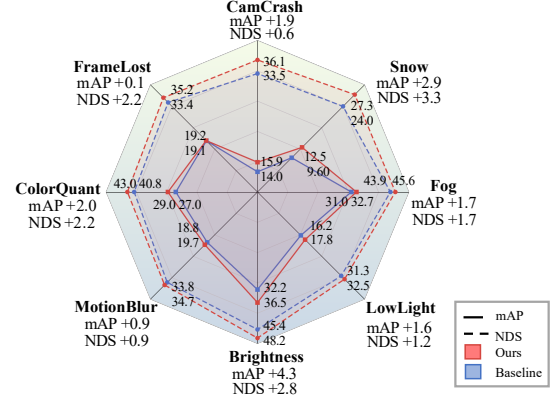


Figure 5. Performance comparison under different corner cases.

our confidence that vision-language multi-modality learning could serve as a promising approach to mitigate feature degradation in real-world driving environments.

4.4. Component-wise Analysis

Different Prototype Settings. In our framework, the introduced multi-modal prototypes provide prior knowledge to compensate the diminished visual features through query initialization and sampling. As shown in Tab. 3, we study the influence of different settings for the prototypes. ① denotes the baseline results (SparseBEV [28]). By exploiting the text prototype for occlusion-aware correlation learning, the performance noticeably improves for 1.1% mAP and 1.5% NDS respectively (② v.s. ①). Applying both text and visual prototypes further brings gains for 0.6% mAP and 0.3% NDS (③ v.s. ②). This demonstrates the effectiveness of our design for 3D detection, as well as the complementarity of visual and language knowledge. Notably, the performance of each visibility level obtains consistent improvements, showing the enhanced capability of our method in detecting objects in different occlusion conditions.

Different Strategies in Occlusion-aware Training. We improve the training strategies of baseline framework by introducing the pseudo-occlusion processor (P2) and multi-modal contrastive learning (CL). Tab. 3 shows that CL brings gains of 0.4% mAP and 0.5% NDS (④ v.s. ③). Applying P2 further improves for 0.6% mAP and 0.3% NDS (⑤ v.s. ④). The performance of each visibility level is also consistently enhanced. Notably, the proposed two training strategies introduce rare training costs and zero inference burden. Meanwhile surprisingly, changing the occlusion distribution of training samples doesn't bring negative effects to the overall performances, which we believe more methods can benefit from this cost-free lunch.

4.5. Further Analysis

Prototype Fine-tuning. As aforementioned, we fine-tune the multi-modal prototypes to adapt the detection task and

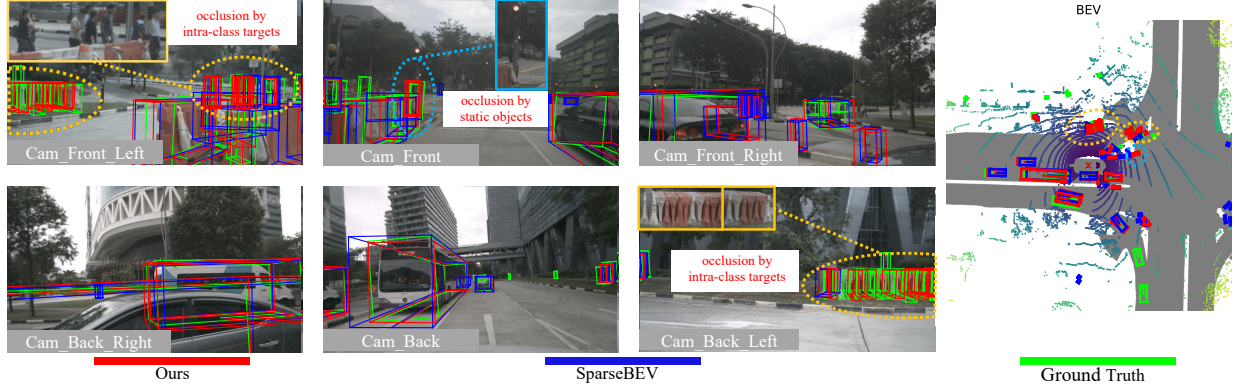


Figure 6. Visualization results of SparseBEV and our CorrBEV_{sp} in two typical occlusion cases (see dashed circles).

our model. Tab. 4 presents the effect of this design. Not surprisingly, if we freeze one of them during training (② and ③), the performance degrades. Compared with language prototypes, freezing the visual one brings more negative effects to the model, where the mAP drops from 47.5% to 46.3% (④vs.②). The underlying reason is that the visual prototypes and the original backbone features belong to the same domain, which contains less complementary information than language from another domain. Hence, the visual prototypes need more effort to adapt the model.

Qualitative Results. Fig. 6 presents a qualitative comparison between the baseline SparseBEV and our proposed CorrBEV_{sp}. As previously discussed, our primary concern is the threat to safe driving caused by occlusion. Therefore in the figure, we highlight two typical occlusion scenarios where our CorrBEV_{sp} demonstrates its positive effect. Firstly, in the “front” camera, a pedestrian partially occluded by a traffic light pole is successfully identified by our method, whereas it is missed by the baseline SparseBEV (highlighted by the cyan circle). This scenario represents a critical corner case in practical driving environments. Missing detection of such objects could lead to collisions, resulting in casualties and property damage. Even late detection could necessitate emergency braking, thereby reducing driving comfort. Secondly, in the “front left” and “back right” cameras, our method more accurately identifies crowded pedestrians and barriers occluded by intra-class objects, while the baseline method misses some of these targets (yellow circles). In summary, the qualitative comparison demonstrates the advances of our method in detecting partially occluded objects. We believe our study will contribute positively to safer autonomous driving.

Contrastive Alignment. One goal of our multi-modal contrastive learner is to increase the separation between different categories and pull the objects in the same category closer in high-dimensional space. “Motorcycles, bicycles and pedestrians” are easily confused types, as a person is frequently riding the bike. Fig. 7 visualizes the hidden features of the above three categories using t-SNE [3]. It

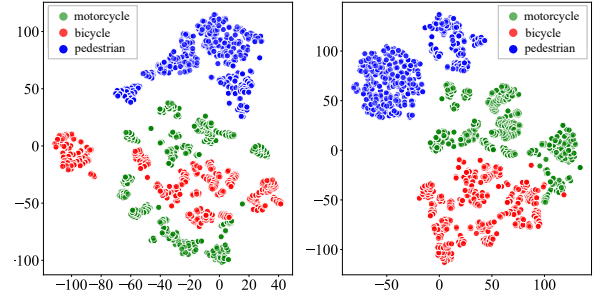


Figure 7. The visualization of embeddings with t-SNE for inter-class (pedestrian, bicycle, motorcycle). Left: our baseline [28], Right: with our contrastive learner.

demonstrates that our contrastive learner effectively compacts the embeddings for objects within the same category, reducing the confusion between bicycles and motorcycles. Type confusion is a significant issue in practical autonomous driving systems because downstream tracking and prediction tasks may assign different motion models to these categories. Therefore, we believe this design not only enhances the perception but also benefits motion prediction.

5. Conclusion

In this paper, we propose CorrBEV, a plug-and-play framework to address the occlusion challenge for camera-based multi-view 3D detection task. The essence is introducing multi-modal prototypes as prior knowledge for occlusion reasoning. By incorporating the well-designed occlusion-aware trainer, we demonstrate consistent improvements in two different baseline frameworks. Besides, we conduct a thorough evaluation and analysis of detection performance under different occlusion conditions, which we believe is an insightful contribution to the field. Notably, our occlusion-aware design methodology aligns with real-world requirements and we expect the community to divert more attention to occlusion handling in BEV perception.

Acknowledgements. This work is supported by the National Natural Science Foundation of China No. 62476235, Hebei Natural Science Foundation No. F2023203012 and Innovation Capability Improvement Plan Project of Hebei Province No. 22567626H. Heng Fan is not supported by any funds for this work.

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